



A Typologically-Motivated Approach to Automatic Morphotactic Inference

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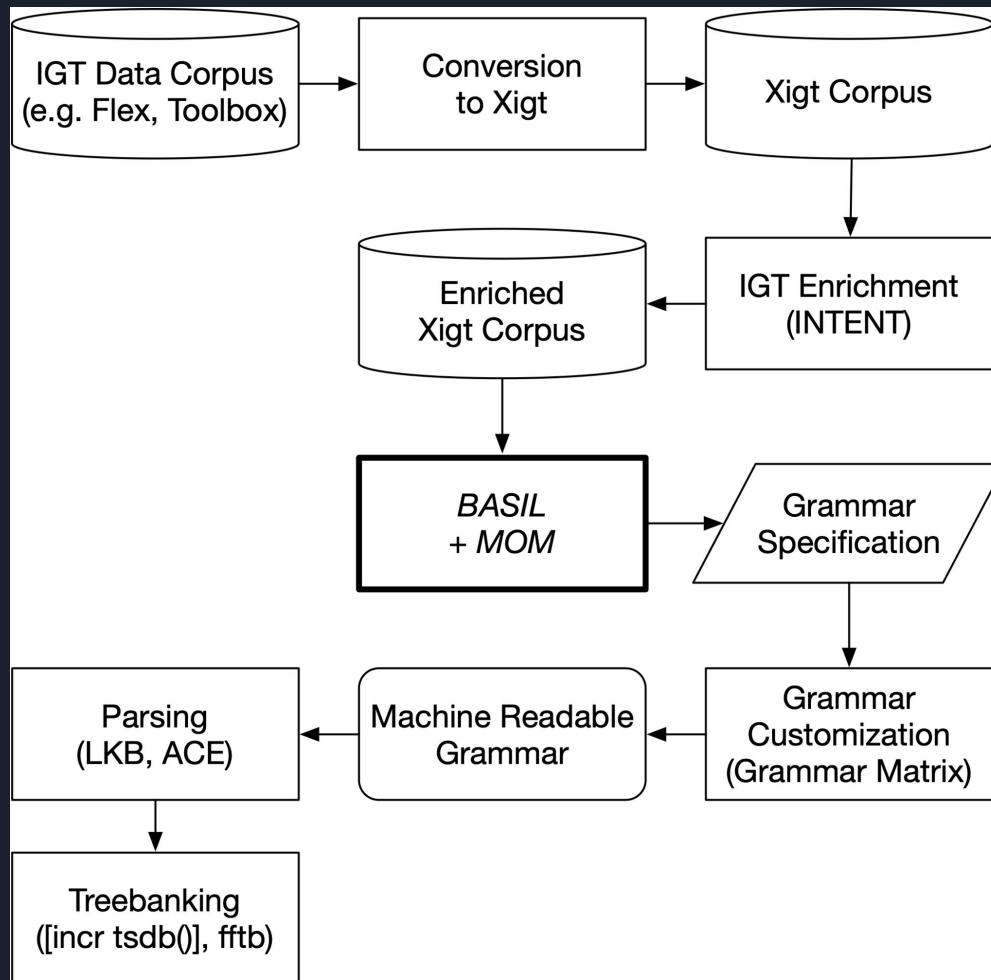


The Grammar Matrix

- Tool for creating implemented grammars
- Core grammar + modular libraries (for e.g. valence-changing morphology, noun incorporation)
- HPSG + MRS
- Interactive web app → choices file

AGGREGATION

- Automatic Generation of Grammars for Endangered Languages from Glosses and Typological Information
- Rule-based grammar inference system
- (Language Documentation →) Interlinear Glossed Text (IGT) → choices file
- IGT is a rich source of information, but only because of the hard work of linguists and community members



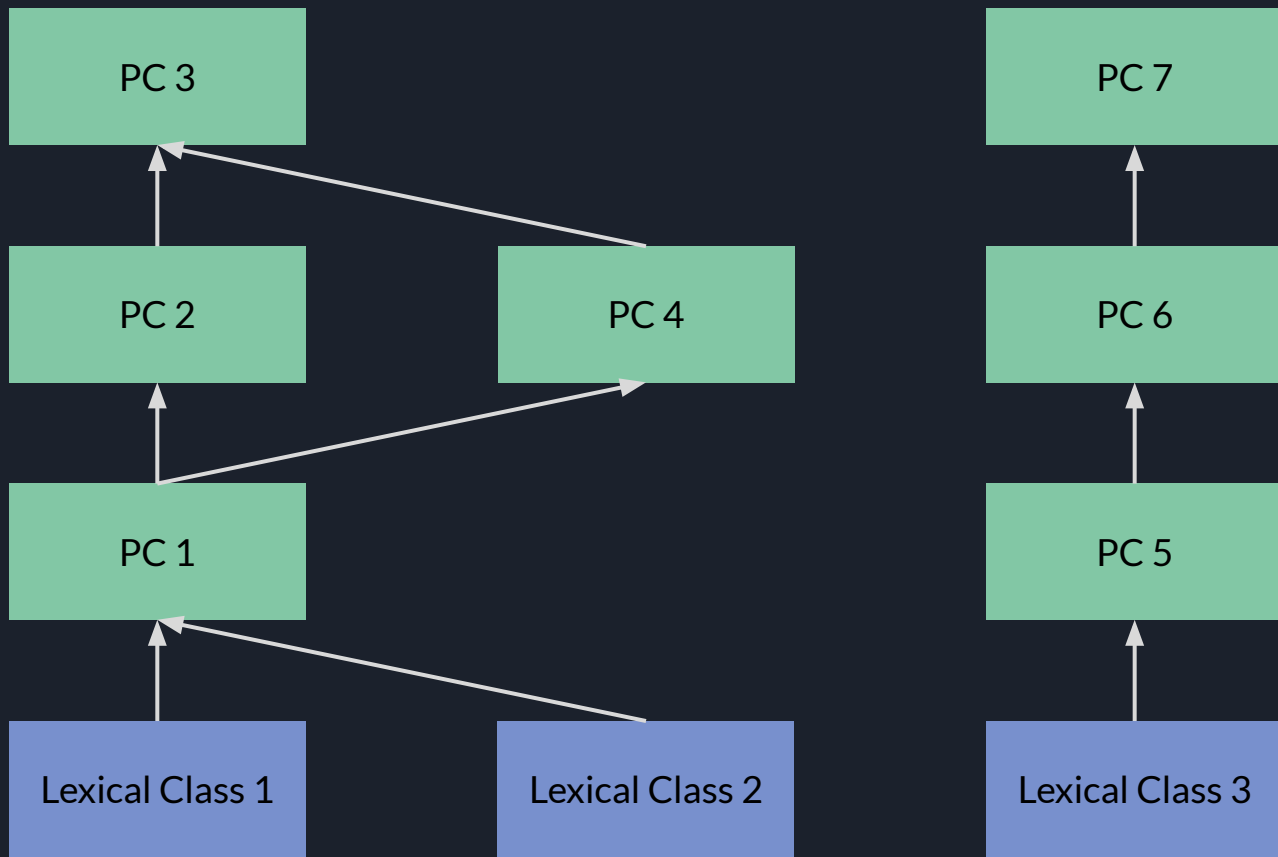


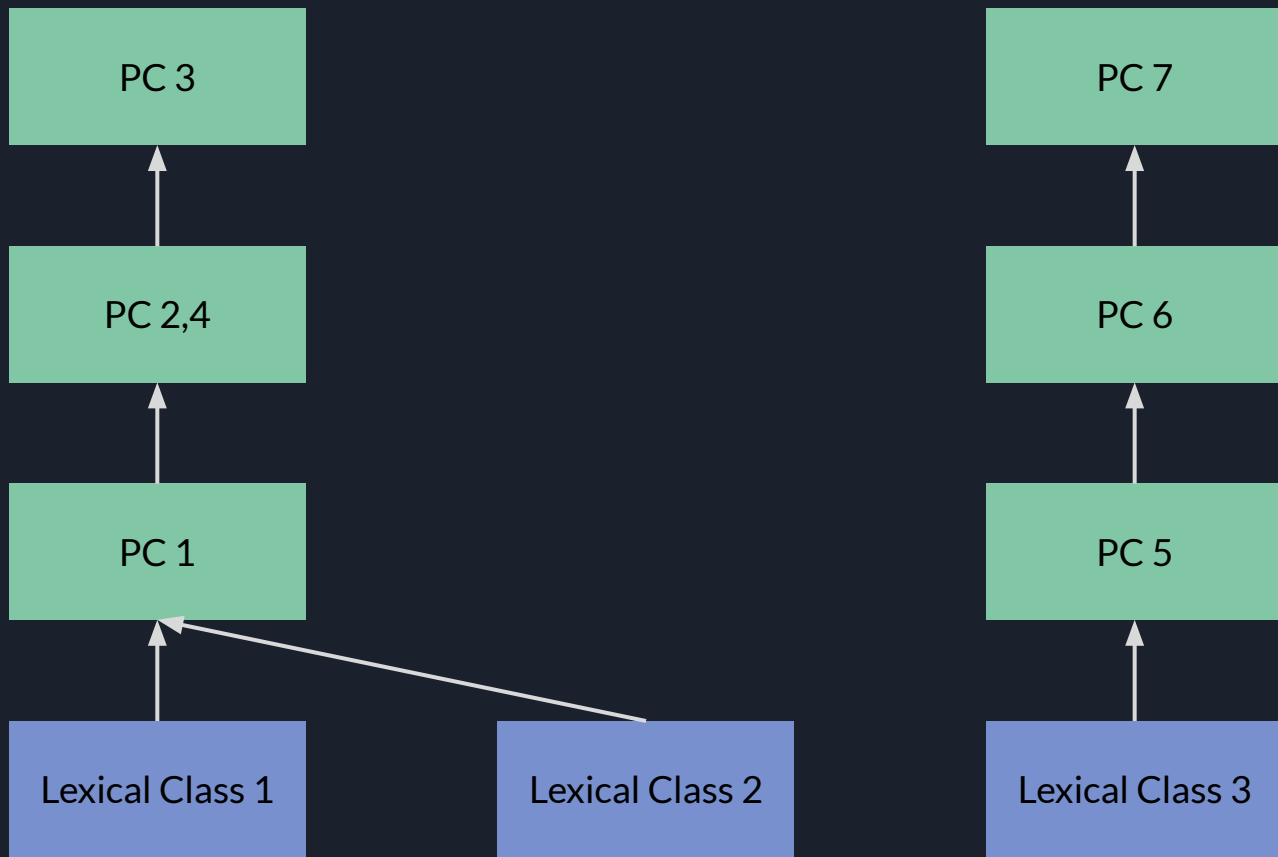
MOM

- Infers the morphotactics of nouns and verbs
- The result of this is a set of lexical classes (LCs) and a set of position classes (PCs)
- Assumes a fixed order of affixation (all prefixes, then all suffixes)

Basic process:

- For each word in the dataset, initialize an LC for the stem and a PC for each affix and add the appropriate inputs
- Rank all pairs of PCs in terms of input overlap
- Merge the pair with the highest overlap and update the affected overlap scores
- Keep merging and updating until no pair of PCs has an overlap value greater than the configured threshold







Research Question

- How can the grammatical categories of grams in IGT gloss lines be utilized to enhance the automatic inference of morphotactics?
- Primary hypothesis: morphotactic inference can be improved by using grammatical feature information to (1) model canonical morphotactics and (2) anticipate instances of common typological patterns
 - Canonical PC: collection of “morphs expressing different values for the same feature cluster in a single linear position, strictly ordered with respect to positions serving for the realisation of other features” (Bonami and Cysmann, 2013, pg. 28)
 - Typological parameters: flexivity, cumulative/fused exponence

Methodology: new overlap function

- Centered around feature overlap
- Input overlap prerequisite
- Input overlap fallback
- Feature cluster boosting
- Configuration change: “unlimited” merging rounds (previously 1)

Algorithm 3 Feature-based overlap function with feature-specific considerations

```
1: function BOOST_OVERLAP(feats1, feats2, feat_set)
2:   if  $|feats1 \cap feat\_set| \geq 1 \wedge |feats2 \cap feat\_set| \geq 1$  then
3:     if  $|feats1 \cap feat\_set| + |feats2 \cap feat\_set| \geq 3$  then
4:        $feats1 \leftarrow feats1 \cup feat\_set$ 
5:        $feats2 \leftarrow feats2 \cup feat\_set$ 
6:   return feats1, feats2
7: function OVERLAP(pc1, pc2)
8:    $input\_overlap \leftarrow |pc1.inputs \cap pc2.inputs| / |pc1.inputs \cup pc2.inputs|$ 
9:   if  $|pc1.feats| = 0 \vee |pc2.feats| = 0$  then ▷ input fallback
10:    return input_overlap
11:   if pc1.pos_tag = 'verb' then ▷ boosted feature groupings
12:      $pc1.feats, pc2.feats \leftarrow BOOST\_OVERLAP(pc1.feats, pc2.feats, \{'P', 'N', 'G'\})$ 
13:      $pc1.feats, pc2.feats \leftarrow BOOST\_OVERLAP(pc1.feats, pc2.feats, \{'T', 'A', 'M'\})$ 
14:   else
15:      $pc1.feats, pc2.feats \leftarrow BOOST\_OVERLAP(pc1.feats, pc2.feats, \{'N', 'case'\})$ 
16:    $feat\_overlap \leftarrow |pc1.feats \cap pc2.feats| / |pc1.feats \cup pc2.feats|$ 
17:   if input_overlap > 0 then ▷ input overlap prerequisite
18:     return feat_overlap
19:   else
20:     return 0
```




Methodology: Gram map

- Additional configuration file where dataset glosses (left column) can be mapped to MOM's internal feature values (right column)
- Aims to reduce the information loss that occurs when the IGT data is ingested (unrecognized glosses are ignored)
- Can also be used to correct misinterpretations, e.g. IMP is mapped to imperfective aspect by default but denotes imperative mood in a particular dataset

GPL	pl
IMP	imp
INTEND	intt
PERMIT	perm
PRO	prosp
PROHB	proh
START	incep



Evaluation Methodology

- Data-driven development: 3 development datasets, 3 test datasets
- Evaluation: 10-fold cross-validation
- Metrics
 - Grammar performance: lexical coverage, sentences parsed, amount of ambiguity
 - Grammar complexity: # LCs, # PCs
- Manual inspection
 - Analyzed with respect to the corresponding dataset, not necessarily the language itself

Language [iso]	Number of IGT	Number of Words	Morphemes per Word	Type Stems Ratio	Allomorph Ratio
Lezgi [lez]	1,531	18,724	2.17	3.05	1.48
Manipuri [mni]	1,785	15,887	1.51	1.80	1.39
Wambaya [wmb]	797	2,933	1.54	1.86	1.35
Wakhi [wbl]	770	3,219	1.79	1.86	1.17
South Efate [erk]	2,240	24,774	1.4	2.55	1.38
Hiaki [yaq]	10,971	77,894	1.19	1.44	3.69

Table 3.1: Dataset characteristics of development and test datasets (Liu, 2025)

Results: Grammar Performance



Dataset	Lexical Coverage	Coverage	Ambiguity	Max Results
Wambaya	8.19 %	1.83 %	26.33	356
Wambaya	8.44 %	1.83 %	26.33	356
Lezgi	11.61 %	9.36 %	3,315.30	28,110
Lezgi	11.11 %	9.11 %	2,087.67	23,968
Manipuri	6.99 %	6.17 %	3,340.79	32,592
Manipuri	7.05 %	6.23 %	1,697.07	23,711
Wakhi	25.77 %	13.47 %	18.21	586
Wakhi	25.18 %	14.79 %	18.68	586
South Efate	14.87 %	7.01 %	5,628.68	20,013
South Efate	14.89 %	6.93 %	5,928.53	21,786
Hiaki	16.47 %	9.71 %	131.47	11,904
Hiaki	16.55 %	9.35 %	30.33	1,125

Table 5.13: Baseline and Final Grammar Performance Metrics by Dataset

Results: Grammar Complexity



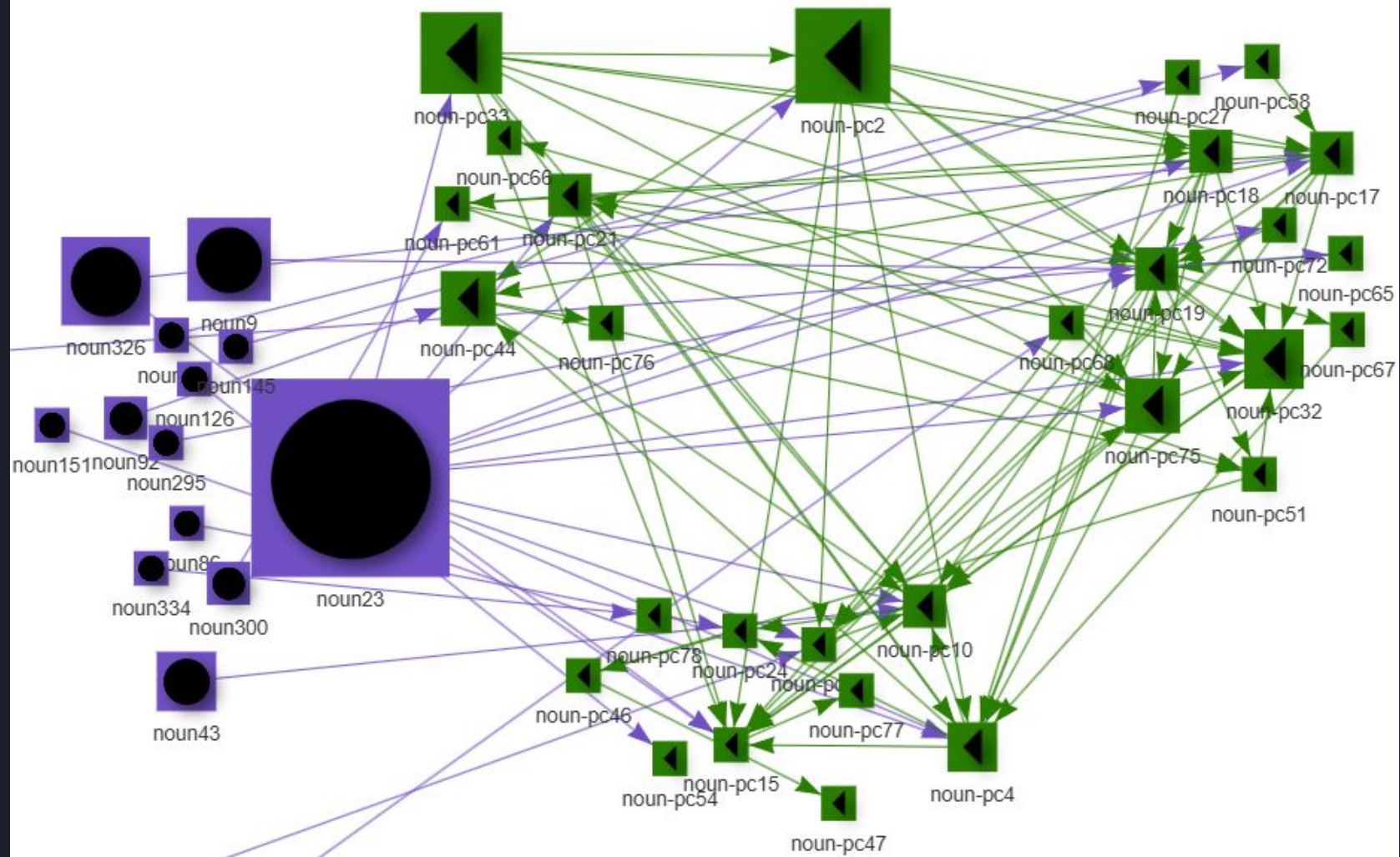
Dataset	N LCs	N PCs	V LCs	V PCs
Wambaya	24.1	20.3	19.3	24.1
Wambaya	23.8	23.1	19.3	24.1
Lezgi	61.2	27.2	41.1	27.0
Lezgi	53.5	26.2	29.1	25.1
Manipuri	20.8	23.7	33.2	32.9
Manipuri	17.7	23.0	25.5	34.1
Wakhi	12.0	6.9	8.6	9.1
Wakhi	11.9	6.9	6.6	9.7
South Efate	34.5	11.9	13.8	11.1
South Efate	20.7	9.0	11.8	9.8
Hiaki	23.7	11.7	34.0	24.7
Hiaki	19.8	11.5	19.5	20.0

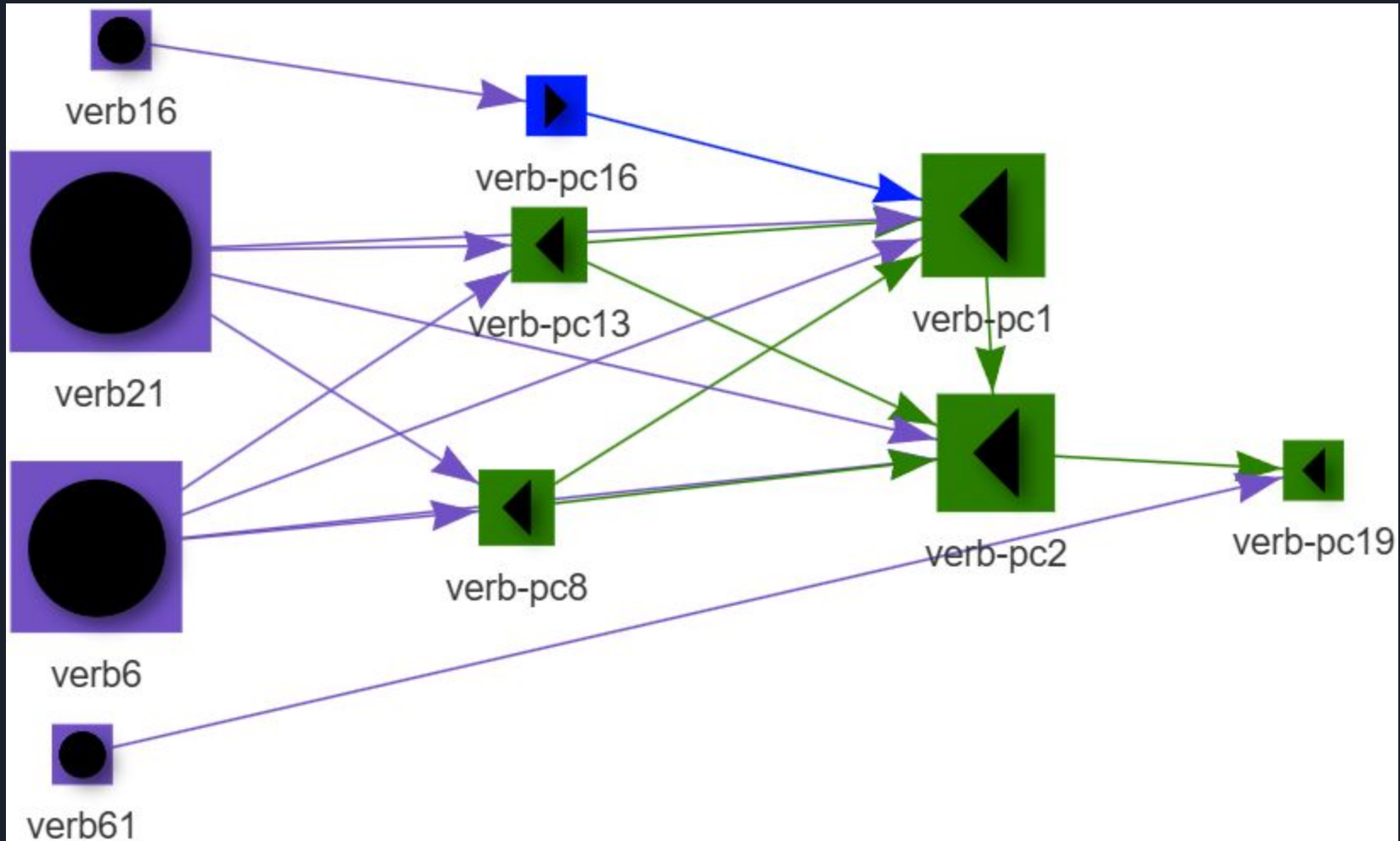
Table 5.14: Baseline and Final Grammar Complexity Metrics by Dataset



Discussion

- Changes had similar effects for the development and test datasets
- Coverage was not affected but the amount of ambiguity decreased in some cases
- Number of LCs/PCs stayed the same or decreased (with a few exceptions)
 - The most significant instances of additional merging were due to the increased number of merging rounds
- PCs were much more canonical on average
- Some inputs graphs remained relatively unchanged from the baseline
 - Featureless affixes are likely the culprit
- Feature cluster boosting had little effect
- Addition of manual gram maps did not affect metrics, but reduced information loss







Future work

- Case study with a specific language
- Prioritize feature overlap over input overlap to a greater extent
- More accurate tracking of the features expressed by affixes in a PC, specifically for those that can be subject- or object-marking
- Use information from the gram map in the rest of the inference algorithm



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